

Brennan Duck

LinkedIn | GitHub | Portfolio | Knoxville, TN

Education

Western Governors University | *B.S. Computer Science* | **Currently Attending**

- Focus: Data Structures, Algorithms, Operating Systems, Computer Architecture

Technical Skills

- **Languages:** Go (Golang), C, C++, Python, SQL, Bash, Java, TypeScript
- **Backend & Systems:** RESTful APIs, Distributed Systems, Concurrency, Memory Management, TCP/IP, Relational Databases
- **Cloud & Infrastructure:** AWS (S3, Lambda, EC2, IAM), Docker, Coolify, Linux/Unix, Git, PostgreSQL

Projects

Deterministic PII Redactor | *Go, Data Structures, REST API* | **April 2026 – Present**

High-performance PHI/PII redaction engine using regex, Aho-Corasick, and English grammatical patterns

- Architected a fully deterministic and auditable PHI/PII redaction pipeline, ensuring compliance and traceability for sensitive healthcare data
- Designed and implemented a custom name detection system combining Aho-Corasick algorithm with grammatical rules, achieving up to 30x faster performance than traditional NER and LLM-based approaches
- Built and deployed a REST API using Chi router to serve the redaction engine efficiently with low latency

Bonobo Language Interpreter | *Go, AST, Parsing* | **Nov 2025 – Present**

A custom interpreted programming language written in Go.

- Implemented a Pratt Parser (Top-Down Operator Precedence) to handle complex expressions without parser generators.
- Built a REPL and AST-dumper to visualize parsing logic and tokenization in real-time.

Professional Experience

ContinuumCloud | *Software Engineering Intern* | **May 2026 – Present**

Welligent AI and innovation team

- Architected and developed a Proof of Concept (PoC) for a modern Behavioral Health EHR system, designing scalable architecture to modernize legacy Welligent workflows and improve clinical documentation efficiency.
- Built an insurance discovery tool that automates patient insurance verification and eligibility checks, streamlining revenue cycle processes within the Welligent platform.
- Developed an internal data extraction tool that automates retrieval and processing of documents from SharePoint, significantly reducing manual data collection time for the AI and Innovation team.

Freelance | *Software Developer* | **May 2024 – Present**

- Developed performant web solutions for local businesses, integrating legacy systems with modern RESTful APIs to automate data entry and reporting workflows.

Slopopedia | *Software Engineering Extern* | **March 2024 – May 2024**

A community platform for cult and B-movie enthusiasts

- Operated in an Agile environment, participating in code reviews and sprint planning to improve development velocity.
- Implemented S3 presigned URL workflows to optimize secure asset delivery for media-heavy pages, reducing load times and eliminating public bucket exposure.